



Electrostatic coupling and water bridging in adsorption hierarchy of biomolecules at water–clay interfaces

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Clay minerals are implicated in the retention of biomolecules within organic matter in many soil environments. Spectroscopic studies have proposed several mechanisms for biomolecule adsorption on clays. Here, we employ molecular dynamics simulations to investigate these mechanisms in hydrated adsorbate conformations of montmorillonite, a smectite-type clay, with ten biomolecules of varying chemistry and structure, including sugars related to cellulose and hemicellulose, lignin-related phenolic acid, and amino acids with different functional groups. Our molecular modeling captures biomolecule–clay and biomolecule–biomolecule interactions that dictate selectivity and competition in adsorption retention and interlayer nanopore trapping, which we determine experimentally by nuclear magnetic resonance (NMR) and X-ray diffraction, respectively. Specific adsorbate structures are important in facilitating the electrostatic attraction and Van der Waals energies underlying the hierarchy in biomolecule adsorption. Stabilized by a network of direct and water-bridged hydrogen bonds, favorable electrostatic interactions drive this hierarchy whereby amino acids with positively charged side chains are preferentially adsorbed on the negatively charged clay surface compared to the sugars and carboxylate-rich aromatics and amino acids. With divalent metal cations, our model adsorbate conformations illustrate hydrated metal cation bridging of carboxylate-containing biomolecules to the clay surface, thus explaining divalent cation-promoted adsorption from our experimental data. Adsorption experiments with a mixture of biomolecules reveal selective inhibition in biomolecule adsorption, which our molecular modeling attributes to electrostatic biomolecule–biomolecule pairing that is more energetically favorable than the biomolecule–clay complex. In sum, our findings highlight chemical and structural features that can inform hypotheses for predicting biomolecule adsorption at water–clay interfaces.

clay minerals | adsorption | soil organic matter | spectroscopy | molecular simulations

Organic matter in soil represents a large terrestrial stock, which is approximately three times greater than the carbon stored in living biomass (1). Soil organic matter includes low-molecular-weight biomolecules, which are susceptible to microbial turnover to contribute to carbon dioxide efflux, but their bioavailability is limited by association with minerals (2). More than 60% of organic carbon in soil is estimated to be bound to minerals (3), facilitating long-term preservation of soil organic carbon (4). Smectite clays, which can constitute up to 50 to 70% of minerals in certain soils (5), include montmorillonite (MMT), illite, beidellite, and nontronite (6). Smectite-type clays present important adsorbents due to their high specific surface area, structural negative charges from isomorphous substitutions, and swelling property that mediates intercalation (7). The negative charges in the smectite layer structure arise from isomorphous substitution in the octahedral sheet (Mg^{2+} for Al^{3+}) or in the tetrahedral sheet (Al^{3+} for Si^{4+}); charge-compensating metal cations in the interlayer space satisfy these structural negative charges (8). Importantly, adsorption within smectite interlayers is implicated in stabilizing organic matter in soils for several thousand years (9). Depending on interaction mechanisms, biomolecules in soils can participate in competitive, non-competitive, or promoted adsorption behavior at clay interfaces (10, 11).

Multiple mechanisms were proposed for the adsorption of biomolecules with different functional groups on smectite clays (12): electrostatic attraction for positively charged organic compounds such as lysine, arginine, and alkylamine (13, 14); cation bridging via multivalent metal cations (Ca^{2+} and Mg^{2+}) between the clay surface and negatively charged biomolecules including nucleic acids, proteins, and other carboxylate-containing compounds (15–19); hydrogen bonding (H-bonding) and other Van der Waals interactions for uncharged polar compounds such as sugars and polysaccharides (20–22). Which of these interaction mechanisms dictates adsorption selectivity remains to be elucidated. For a natural organic matter assemblage reacted with MMT, there was preferential adsorption

Significance

The storage of biomolecules derived from plants and microbes in soils represents an important reservoir at the interface of carbon sequestration and biogeochemical carbon cycling. Association with clay minerals controls the residence time of these biomolecules in soils in semi-arid and temperate climates, by influencing accessibility to microbial turnover. Here, we combine experiments and molecular modeling to investigate biomolecule–clay complexes serving as precursors to soil organic matter. We obtain adsorbate structures and binding energies to reveal electrostatic and water-bridging interactions that drive the adsorption and nanopore trapping of chemically diverse biomolecules within smectite-type clays. These findings advance a fundamental understanding of the mechanisms critical to the role of clays in organic matter retention in different soil conditions.

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of hydrophilic components containing amino acid-, phenol-, and carbohydrate-like moieties compared to hydrophobic components such as lipids and condensed aromatics (23). Moreover, adsorption of phenolic acids such as salicylate was shown to be inhibited by the presence of amino acids (10). And, desorption of previously bound low-molecular-weight aromatics was observed following the addition of natural organic matter (24). These previous findings implied selective adsorption of biomolecules with specific functional groups on the clay surface, but the mechanisms that drive this hierarchical adsorption remain unclear.

Here, we combined adsorption experiments with molecular modeling simulations to investigate ten biomolecules (seven amino acids, two sugars, and one lignin derivative) reacted with MMT. These biomolecules represent chemical diversity in soil organic matter precursors: the two sugars (glucose, a six-carbon sugar; xylose, a five-carbon sugar), which are uncharged polar compounds, are monomers from the abundant plant biopolymers of cellulose and hemicellulose, respectively (25, 26); the phenolic acid (*p*-coumarate) represents common aromatic components in the lignin polymer (25, 26); and, the amino acids have positively charged (lysine, histidine), negatively charged (glutamate), uncharged polar (glutamine, threonine, histidine), uncharged nonpolar alkyl (leucine), or uncharged aromatic ring (phenylalanine) side chains (Fig. 1A). Using liquid-state ^1H nuclear magnetic resonance (NMR) for solution analysis (Fig. 1A), we conducted adsorption isotherm experiments at pH 7.0 to determine the adsorption distribution coefficient (K_d) for each biomolecule as a single compound or in a mixture, with either Na^+ or Mg^{2+} as the charge-compensating cation, henceforth referred to as Na-MMT and Mg-MMT systems, respectively. We interpreted our results in relation to reported Fourier-transform infrared (FTIR) and solid-state NMR spectroscopic data used to propose mechanisms in clay-organic interactions (23, 27, 28). Molecular modeling simulations are widely used to obtain structural insights into adsorption mechanisms (27, 29), especially for hydrated systems that are challenging for most spectroscopic analyses. We employed a Monte Carlo modeling approach with molecular dynamics equilibration to probe the adsorption mechanisms and interaction energies in the biomolecule–clay adsorbate structures to explain the experimentally determined hierarchy in biomolecule adsorption. The findings from the present study provide structural insights to advance mechanistic underpinnings of biomolecule–clay and biomolecule–biomolecule interactions that drive the fate of freshly generated biomolecules at clay interfaces in soil and sediment matrices.

Results

Electrostatic Attraction Energy Explains Hierarchy in the Adsorption Affinity of Structurally Different Biomolecules to the Clay. The adsorption isotherms revealed a hierarchy in the adsorption affinity of the ten biomolecules, each as a single adsorptive, on MMT (Fig. 1B and *SI Appendix, Figs. S1–S4*). With Na-MMT, the lowest K_d values were for two negatively charged biomolecules (on average, 0.26 L kg^{-1} for glutamate and 0.04 L kg^{-1} for *p*-coumarate) (Fig. 1B), consistent with reported poor adsorption affinity for related compounds (30, 31). In contrast, the K_d values for lysine (16.4 L kg^{-1}) and histidine (3.4 L kg^{-1}) with Na-MMT were about 3 to 200 times the values obtained with the other biomolecules (Fig. 1B). To determine the extent of interlayer adsorption, we monitored the expansion of layer-to-layer distance (d_{001}) using X-ray diffraction data (Fig. 1C). For the biomolecules with measurable adsorption with Na-MMT, there was an increase in d_{001} as a function of amount of the adsorbed biomolecules, in agreement with previous findings (Fig. 1C) (32).

Furthermore, we found that the extent of intercalation within the clay layers was correlated positively with the adsorption hierarchy reflected by the K_d values, implying that the interlayer trapping of the biomolecules facilitated their adsorption retention (Fig. 1C).

In accordance with spectroscopic evidence (27, 30), the model adsorbates with the amino acids exhibited the positively charged amino group were generally closer (2.7 to 4.8 Å from the N atom) than the negatively charged carboxyl group (3.4 to 6.6 Å from the C atom) to the Na-MMT surface (Fig. 1D). Notably, the binding conformations for lysine (with its two positively charged amino groups) and histidine (with its positively charged amino group and imidazole group) facilitated up to threefold higher electrostatic attraction energy ($P < 0.001$), compared to the other adsorbates (Fig. 1D and E). Adsorption of lysine to MMT has been extensively studied (27, 33–37), but there were discrepancies in the proposed binding mechanism whereby either the side-chain amino group (33, 35) or the alpha amino group (36) of lysine was proposed to drive the binding of lysine to the MMT surface. However, we found that both amino groups were involved in anchoring the adsorbed lysine to the mineral interface to satisfy two negatively charged sites on the MMT surface (Fig. 1D), thus leading to the relatively high electrostatic attraction energy of the lysine–MMT complex compared to the other biomolecule–clay complexes (Fig. 1E). With regards to histidine, we obtained two H-bonding interactions between the secondary amino group in the imidazole ring and exposed oxygen atoms on the MMT siloxane surface, in addition to the positioning of the positively charged alpha amino group to charge balance a negatively charged cavity on the MMT (Fig. 1D).

The trend in the electrostatic attraction energy of the different model biomolecule–MMT complexes generally correlated positively with the K_d values obtained experimentally during the adsorption experiments, thus further highlighting the importance of favorable electrostatic interactions (Fig. 1D and E). The molecular modeling also predicted the contribution of water-bridged and direct H-bonds in facilitating the adsorption of all investigated biomolecules (Fig. 1D). Due to the co-occurrence of these different H-bonds and their simultaneous contributions to the Van der Waals energies, it is challenging to disentangle the specific contribution of water-bridging interactions. On one hand, for poorly adsorbed carbohydrates (xylose and glucose) that lack a functional group to be involved in favorable electrostatic interactions but were found to be involved in a network of direct and indirect H-bonds, their model adsorbates had the lowest values in electrostatic attraction energies that were similar to the Van der Waals energies (Fig. 1B and E). On the other hand, for the model adsorbates (lysine and histidine) with the highest K_d values and with functional groups that can mediate both electrostatic and H-bonding interactions on the MMT surface, the electrostatic attraction energies were up to about fourfold higher than the corresponding Van der Waals energies ($P < 0.001$) (Fig. 1B and E). These findings thus implied that H-bonding interactions alone were insufficient to mediate strong adsorption. Taken collectively, our molecular modeling results led us to conclude that electrostatic attraction dictated the adsorption hierarchy on the MMT surface and that the H-bonding interactions were important in stabilizing the adsorbate conformations.

Importance of Water-Bridging Interactions in Facilitating Favorable Adsorption of Biomolecules. To probe further the possible role of water bridging in the adsorption process, we followed the diffusion trajectory of a biomolecule from being initially localized in the bulk solution to the eventual adsorption at the Na-MMT surface, focusing on the two biomolecules (lysine

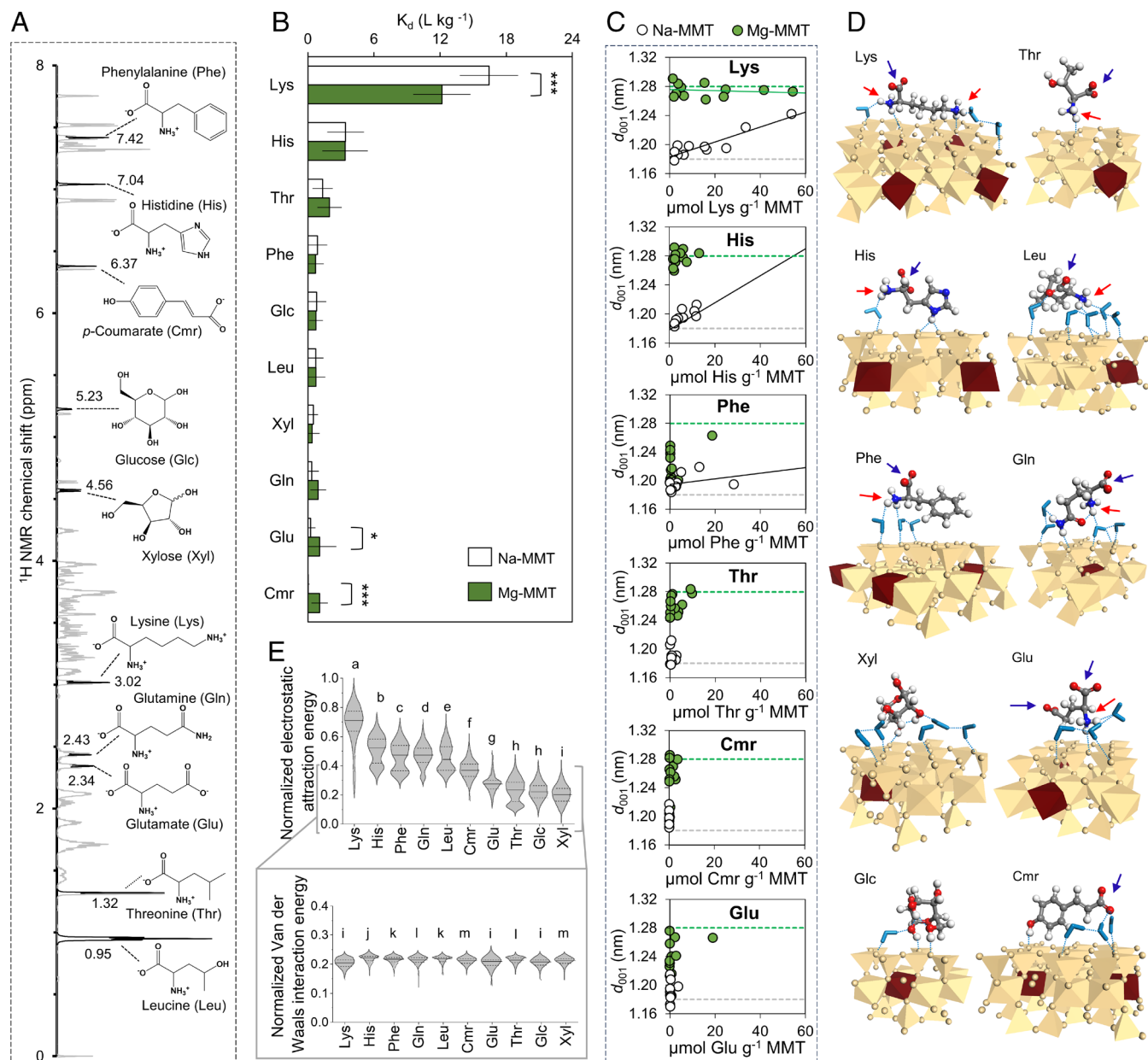


Fig. 1. Adsorption hierarchy and binding conformations of biomolecules on a smectite clay. (A) Chemical structures and liquid-state ¹H NMR spectrum for the ten investigated biomolecules; bold portion in the NMR spectrum with the specific ppm value indicates peak used for each biomolecule quantitation; chemical structure is shown at the dominant protonation state at pH 7.0. (B) K_d of each biomolecule in a single-compound solution reacted with Na-MMT (white bars) or Mg-MMT (green bars); a significant difference between the two adsorption scenarios was determined from two-tailed Student's *t* test and denoted by * ($P < 0.05$), ** ($P < 0.01$), and *** ($P < 0.001$). (C) X-ray diffraction-determined interlayer nanopore size (d_{001}) in Na-MMT (white symbols) and Mg-MMT (green symbols) as a function of adsorbed amount of selected biomolecules ($\mu\text{mol biomolecule g}^{-1}$ of MMT); the solid lines are intended as eye guides for increasing d_{001} values; dashed lines represent the d_{001} of the reference Na-MMT (dashed gray line) and reference Mg-MMT (dashed green line). (D) Optimized model adsorbate structures of the ten biomolecules adsorbed on Na-MMT. (E) Normalized energy for (Top) electrostatic attraction and (Bottom) Van der Waals interactions in the in the dynamic configurations of model adsorbates over the course of the molecular dynamics simulations with Na-MMT ($n = 2,000$) where median value is indicated by the horizontal solid line, and first and third quartiles are indicated by the horizontal dashed lines; statistically significant differences ($P < 0.05$) are denoted by a change in letter; significance was determined by two-tailed Student's *t* test. Color scheme for atoms: C (gray), H (white), O (red), N (dark blue), Mg (green), water (light blue), and atoms in MMT (light yellow); isomorphous substitution sites in MMT octahedral sheet ($\text{Al}^{3+} \rightarrow \text{Mg}^{2+}$) in dark brown; red and blue arrows point to positively and negatively charged moieties in the biomolecule, respectively; and, H-bonds and metal coordination are indicated by dashed blue and red lines, respectively. In (D), only water molecules involved in water-bridged H-bonds between the biomolecule and the clay surface are shown; all other water molecules in the hydrated adsorbate conformation are removed for clarity. The H-bond criteria were set to 2.5 Å for the maximum distance and 120° for the minimum angle between the H-bond donor and acceptor. Data for (B), (C), and (E) are provided in *SI Appendix, Table S1–S3*, respectively. Compound name abbreviations are provided in (A) and other abbreviations are described in the text.

and histidine) that exhibited the highest K_d values, (Fig. 2A and *SI Appendix, Fig. S5*). Interestingly, despite the clear adsorption hierarchy determined experimentally for these biomolecules, we observed continuous adsorption and desorption dynamics in the simulated trajectories for all compounds (*SI Appendix, Fig. S5*), thus highlighting the adsorption process as a partitioning

equilibrium instead of an irreversible binding process (38). We monitored the conformations of the biomolecule adsorptives at different adsorption regions with respect to the MMT surface (Fig. 2B). A network of water-mediated H-bonds facilitated the trajectory of both lysine and histidine from the diffuse swarm toward outer-sphere (i.e., water-bridged) complexation and

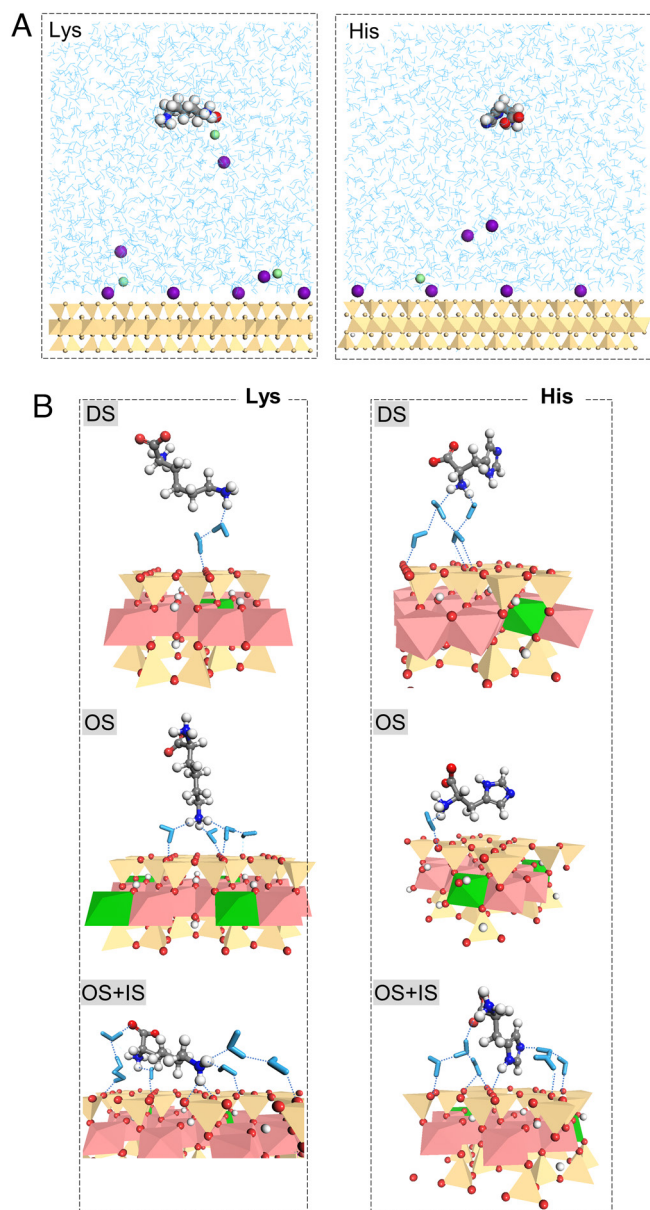


Fig. 2. Dynamics of biomolecule adsorptive diffusion from the bulk solution to the water–clay interface. (A) Initial configurations of lysine and histidine placed in the bulk water, about 2.5 nm from the Na-MMT surface, prior to the molecular dynamics simulations. (B) Configurations of the (Left) lysine (Lys) and (Right) histidine (His) adsorptives in different adsorption zones of Na-MMT: diffusion swarm (DS), outer sphere (OS), and inner sphere (IS) complexation interactions. In (A), all the clay atoms are shown in light yellow for simplicity. Color scheme for the clay atoms in (B): Si (light yellow), H (white), O (red), Al (pink), and Mg (green). Color scheme for biomolecules, waters, and solutes in (A) and (B): C (gray), H (white), O (red), N (dark blue), Na (purple), and Cl (light green), water molecules (light blue). The H-bonds and metal coordination are indicated by dashed blue lines and dashed red lines, respectively. In (B), only water molecules involved in water-bridged H-bonds between the biomolecule and the clay surface are shown; all other water molecules in the hydrated adsorbate conformation are removed for clarity. The H-bond criteria were set to 2.5 Å for the maximum distance and 120° for the minimum angle between the H-bond donor and acceptor. Compound name abbreviations are provided in Fig. 1A.

eventually inner-sphere (i.e., direct) complexation on the MMT surface (Fig. 2B). In this latter conformation, the adsorbed biomolecules participated in direct H-bonds on the clay surface, in addition to water-bridged H-bonds (Fig. 2B). The outer-sphere complexation predicted here by our molecular modeling for the diffusing lysine adsorptive at the water–MMT interface was proposed previously, based on no change in the vibrational

bands in FTIR spectra of lysine in solution versus lysine adsorbed (32). However, monitoring of the water OH stretching band by FTIR spectroscopy has implied the possible displacement of water molecules at the water–clay interface by the lysine adsorptive (34), consistent with our modeling of the changing conformations of adsorbed lysine at the water–mineral interface. In accordance with a previous proposal (33), we also show here that the side-chain amino group in lysine was involved initially in directing the adsorptive to the MMT surface via water bridging, but both the side chain and alpha amino groups participated in the eventual adsorbate conformation (Fig. 2B). The dynamic monitoring of the adsorption process thus emphasized the important role of the extensive network of water-bridging interactions for both the diffusion of the adsorptive toward the mineral surface and the stabilization of the adsorbate complex (Fig. 2B).

Divalent Cations Inhibit Adsorption of Positively Charged Lysine but Promote Adsorption of Glutamate and *p*-Coumarate. For the three biomolecules (lysine, histidine, and threonine) exhibiting the highest K_d values with Na-MMT, the corresponding K_d value with Mg-MMT remained unchanged for both histidine and threonine ($P > 0.10$) but was decreased by 26% for lysine ($P < 0.001$) (Fig. 1B). Competition of metal cations with lysine was proposed previously (32, 39), and our data here further implied that multivalent cations would lead to greater inhibition of lysine adsorption than monovalent cations. Our results also suggested a weak or no competition of histidine and threonine adsorption by the divalent metal cation.

For the two biomolecules (glutamate and *p*-coumarate) with the lowest K_d values with Na-MMT, reaction with Mg-MMT led to a fourfold ($P < 0.05$) and a 27-fold increase ($P < 0.001$) in K_d for glutamate and *p*-coumarate, respectively (Fig. 1B). Interestingly, when Mg^{2+} ions were introduced simultaneously with these two biomolecules as adsorptives for the hydrated clay surface, the molecular simulation results revealed that the carboxylate moiety in both glutamate and *p*-coumarate chelated Mg^{2+} involved in water-bridged outer sphere complexation with the clay surface, thus leading to the enhanced adsorption observed experimentally (Fig. 3). A previous work on promoted adsorption of a carboxylate-containing organic contaminant in the presence of Ca-MMT and Mg-MMT compared to Na-MMT also proposed the occurrence of an outer-sphere complexation of the bound divalent metal cation in the ternary organic-divalent cation–clay complex due to the lack of evidence of inner-sphere complexation based on FTIR and NMR data (40).

The X-ray diffraction data showed that the greater expansion of the interlayers of the reference Mg-MMT (~1.28 nm) than the reference Na-MMT (~1.18 nm) was sufficient for the adsorption of all the biomolecules without further expansion (Fig. 1C and *SI Appendix*, Fig. S6). Moreover, in some cases, a decrease in d_{001} (up to 6.3%) was observed in Mg-MMT populated with adsorbed biomolecules (Fig. 1C); a similar phenomenon was also reported previously with adsorbed organic compounds on Ca-MMT (27). It is acknowledged that the extent of smectite layer expansion would depend on the type of charge-compensating metal cations (41–48), but, here, we attributed the d_{001} decrease to the change in the conformation of the adsorbed biomolecules with Mg-MMT. For instance, compared to the *p*-coumarate configuration with Na-MMT (Fig. 1D), which had the aromatic ring perpendicular to the MMT surface, the Mg^{2+} complexation by the carboxylate moiety of *p*-coumarate facilitated a parallel orientation with respect to the Mg-MMT surface (Fig. 3B). This change in conformation could induce a lower expansion of the MMT interlayer space populated with the biomolecule than would be found in the presence

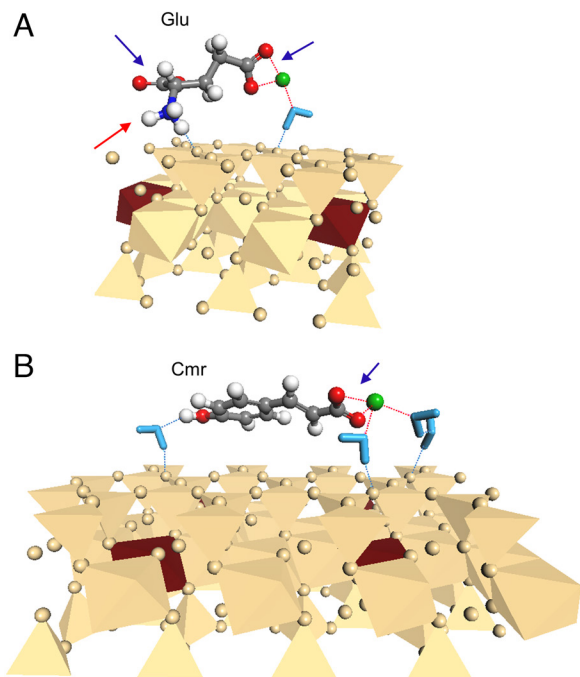


Fig. 3. Divalent metal cation-bridged complexation of biomolecules on the clay surface. Model adsorbate structures of (A) Glu and (B) Cmr with Mg-MMT. Color scheme for atoms: C (gray), H (white), O (red), N (dark blue), Mg (green), water (light blue), and atoms in MMT (light yellow); isomorphous substitution sites in MMT octahedral sheet ($\text{Al}^{3+} \rightarrow \text{Mg}^{2+}$) in dark brown; red and blue arrows point to positively and negatively charged moieties in the biomolecule, respectively, and, H-bonds and metal coordination are indicated by dashed blue lines and dashed red lines, respectively. In the model configurations, only water molecules involved in water-bridged H-bonds between the biomolecule and the clay surface are shown; all other water molecules in the hydrated adsorbate conformation are removed for clarity. The H-bond criteria were set to 2.5 Å for the maximum distance and 120° for the minimum angle between the hydrogen-bond donor and acceptor. Compound name abbreviations are provided in Fig. 1A.

of the fully hydrated Mg^{2+} ion. Previously, the substitution of a divalent metal cation for an adsorbed biomolecule was proposed to be responsible for the decrease in d_{001} (30). In sum, highly adsorbed biomolecules such as lysine and biomolecules such as glutamate with promoted adsorption on Mg-MMT both did not lead to increased expansion of the MMT interlayers, thereby implying the initial interlayer space of the Mg-MMT system was able to accommodate the adsorbed biomolecules on MMT.

Electrostatic Biomolecule–Biomolecule Pairing Inhibits Favorable Adsorption. We investigated to what extent the hierarchy of adsorption affinity determined from the single-compound experiments would remain when the biomolecules were present in a mixture to represent organic assemblages in environmental matrices. To this end, we conducted experiments with all 10 biomolecules present simultaneously in the adsorptive mixture reacted with Na-MMT or Mg-MMT (SI Appendix, Figs. S7 and S8). We obtained the K_d of each compound in the mixture and compared it to the corresponding K_d in the aforementioned single-compound experiments (Fig. 4). In the mixture, there was no change in the K_d values ($P > 0.16$) for the sugars and most amino acids containing either a negatively charged or uncharged side chain, all of which had low adsorption in the single-adsorptive experiments (Fig. 4B). For lysine, the K_d remained unchanged in the mixture with Na-MMT ($P = 0.12$), indicating that lysine adsorption was not impaired by the other nine biomolecules in the mixture (Fig. 4B). However, with Mg-MMT, lysine adsorption was decreased by 27% in the mixture compared to lysine alone

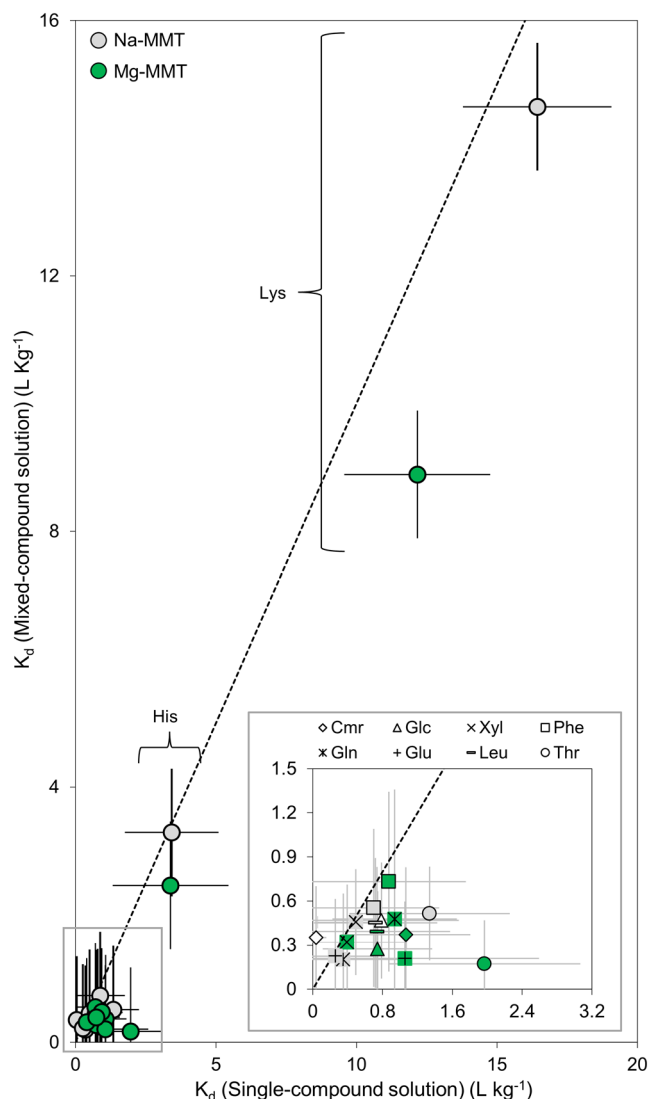


Fig. 4. Adsorption hierarchy of the biomolecules from a mixed-compound solution versus a single-compound solution reacted with Na-MMT or Mg-MMT. Plotted is the K_d value of each biomolecule when present in a mixture with all 10 biomolecules versus the K_d value of the same biomolecule in a single-compound solution reacted with Na-MMT (gray symbols) or Mg-MMT (green symbols). Compound name abbreviations are provided Fig. 1A. Data are provided in SI Appendix, Tables S1 and S4.

with Mg-MMT ($P < 0.001$) (Fig. 4B). Interestingly, the K_d values for glutamate and *p*-coumarate in this same mixture with Mg-MMT were also decreased by 80% ($P < 0.05$) and 65% ($P < 0.01$), respectively, relative to their corresponding single-compound adsorption scenarios (Fig. 4B).

To probe further this seeming competitive adsorption behavior in the mixture reaction with Mg-MMT, we compared K_d values in the following two-compound adsorption scenarios with Mg-MMT: lysine and histidine, lysine and threonine, lysine and glutamate, and lysine and *p*-coumarate (Fig. 5A). Histidine and threonine were chosen because both of these biomolecules had the highest K_d values after lysine (Fig. 1B). Interestingly, when lysine was co-adsorbed with histidine or threonine, there was no change in the K_d of lysine ($P = 0.086$ and $P = 0.104$, respectively) compared to the K_d for lysine alone (Fig. 5A). By contrast, there was a decrease in the K_d for lysine by 27% and 37% when co-adsorbed with *p*-coumarate ($P < 0.001$) or glutamate ($P < 0.001$), respectively; the K_d values for *p*-coumarate and glutamate were also decreased by 20% ($P = 0.50$) and 80% ($P < 0.05$), respectively, relative to

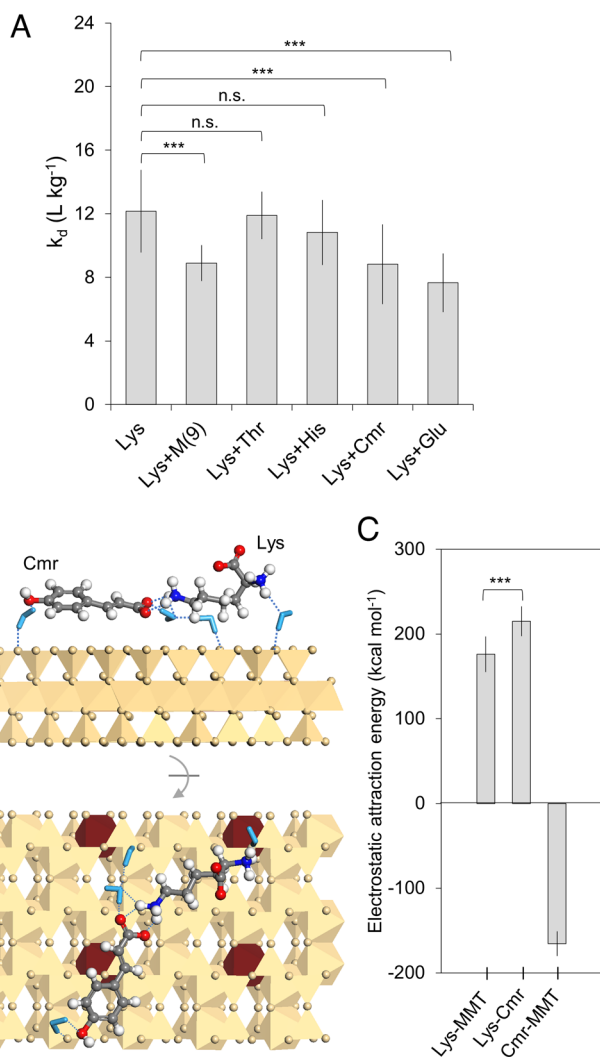


Fig. 5. Competitive adsorption between biomolecules. (A) K_d for Lys alone (Lys) or with nine biomolecules [Lys+M(9)], with threonine (Lys+Thr), with histidine (Lys+His), with Cmr (Lys+Cmr), or with glutamate (Lys+Glu) on Mg-MMT. (B) Model adsorbate structures for co-adsorbed Lys and Cmr on Mg-MMT. (C) Electrostatic attraction energy of biomolecule–clay interaction (Lys-MMT and Cmr-MMT) and biomolecule–biomolecule interaction (Lys-Cmr) in the model adsorbates shown in (B). In (A) and (C), a significant difference was determined from two-tailed Student's *t* test and denoted by *** ($P < 0.001$). Color scheme for atoms in (B): C (gray), H (white), O (red), N (dark blue), Mg (green), water (blue), and atoms in MMT (light yellow); isomorphic substitution sites in MMT octahedral sheet ($Al^{3+} \rightarrow Mg^{2+}$) in dark brown; red and blue arrows point to positively and negatively charged moieties in the biomolecule, respectively; and, H-bonds and metal coordination are indicated by dashed blue and red lines, respectively. In (B), only water molecules involved in water-bridged H-bonds between the biomolecule and the clay surface are shown; all other water molecules in the hydrated adsorbate conformation are removed for clarity. The H-bond criteria were set to 2.5 Å for the maximum distance and 120° for the minimum angle between the H-bond donor and acceptor. Data for (A) and (C) are provided in *SI Appendix, Tables S5 and S6*, respectively.

their corresponding single-adsorptive K_d values (*SI Appendix, Fig. S9*). In separate experiments, we confirmed that a mixture with the remaining five compounds (leucine, phenylalanine, glutamine, glucose, and xylose) had no influence on lysine adsorption ($P = 0.58$) (*SI Appendix, Fig. S10*). Collectively, these data confirmed competitive adsorption between lysine and glutamate, and between lysine and *p*-coumarate.

We conducted molecular dynamics simulations to probe the possible mechanism underlying the inhibition of lysine adsorption on Mg-MMT by *p*-coumarate (*SI Appendix, Fig. S11*). The simulated adsorbate configurations of this co-adsorption revealed

electrostatic coupling between the negatively charged carboxyl group in *p*-coumarate and the positively charged side-chain amino group in lysine (Fig. 4B). In this conformation, the *p*-coumarate-lysine interaction exhibited 22% higher ($P < 0.001$) electrostatic attraction energy than the lysine–MMT complex, which in turn was 16-fold lower ($P < 0.001$) than for the aforementioned lysine–MMT complex in the absence of *p*-coumarate (Figs. 1E and 4C). These findings implied that electrostatic pairing with biomolecules bearing negatively charged carboxylate moieties could undermine the adsorption affinity of positively charged biomolecules to the MMT surface.

Discussion

The retention of biomolecules within smectite-type clay minerals is implicated in facilitating long-term preservation of organic matter in soils, which in turn influences ecosystem-scale carbon cycling (4, 49, 50). Our findings demonstrated that adsorption hierarchy was primarily due to electrostatic attraction in the biomolecule–clay interactions at the water–clay interfaces. Both H-bonding and Van der Waals interactions were also proposed in clay–organic interactions, in particular for uncharged polar compounds such as carbohydrates (28). Notably, we found that water-bridged interactions play an important role in directing the adsorptive from bulk solution to the water–clay interface whereby both water-bridged and direct H-bonding stabilized the binding of all biomolecules on the clay surface. However, the energy profiles of the model adsorbates combined with the adsorption experiment results indicated that these H-bonds alone were not sufficient to explain the hierarchy adsorption with MMT. However, we posit that Van der Waals interactions may play a relatively greater contribution in mediating biomolecule adsorption hierarchy on clays with uncharged basal surfaces such as kaolinite (12).

Consistent with the importance of electrostatic attraction in biomolecule adsorption with smectite clay, cation bridging via common divalent metal cations (Ca^{2+} and Mg^{2+}) in soils has been widely implicated in promoting interactions between negatively charged compounds and smectite clays (17, 51–55). Here, our adsorption experiments also revealed that divalent cations led to enhanced adsorption of carboxylate-bearing biomolecules. Our subsequent molecular modeling predicted that this divalent cation-promoted adsorption involved outer-sphere metal-bridged interactions between the carboxylate moiety and negatively charged sites on the clay surface. This outer-sphere complexation in the metal-bridged ternary complex was proposed previously based on spectroscopic evidence (40).

Regarding competitive adsorption, a previous study has highlighted the preferential adsorption of low-molecular-weight organic matter with a high oxygen-to-carbon ratio onto MMT (53). The specific moieties determined to be adsorbed were protein-like, carbohydrate-like, aromatic-like components whereby their binding is proposed to be facilitated by metal-bridging (53). Our single-adsorptive experiments revealed a preference for protein-related moieties (i.e., certain amino acids) and the importance of metal-bridging in facilitating adsorption of a lignin-related compound (i.e., *p*-coumarate). For the adsorption of a mixture of biomolecules, we found that electrostatic coupling in biomolecule–biomolecule interactions explained competitive adsorption. A similar mechanism was proposed to interfere with the adsorption of natural organic matter of high molecular weight (53). Our modeling data implied that favorable electrostatic pairing between organic matter components may interfere with their expected adsorption affinity to the clay surface. For organic–organic interactions, force microscopy revealed that electrostatic attraction was

up to threefold more thermodynamically favorable than hydrophobic interaction and H-bonding (56), further corroborating a role of intermolecular electrostatic interactions in impeding adsorption affinity as demonstrated here.

With respect to trapping of biomolecules in smectite clay interlayers, previous X-ray diffraction data have demonstrated interlayer adsorption for amino acids, short-chain carboxylic acids, phenolic acids, peptides, urea, and fulvic acids within MMT (30, 57–64). Our data of positive correlation between increased intercalation as a function of an increase in adsorbed biomolecules led us to conclude that intercalation within the clay layers promoted adsorption retention, thus explaining preservation of biomolecules trapped in smectite interlayers (65). Interlayer adsorption of biomolecules precludes accessibility to soil microbes and their extracellular enzymes (66, 67). Therefore, understanding of the selectivity in interlayer adsorption is important to predicting the fate of biomolecules. In sum, the structural insights presented here can guide in formulating hypotheses regarding biomolecule adsorption behaviors on different clay mineral types that play a critical role in the biological cycling versus physical persistence of organic matter in soils and sediments.

Summary of Materials and Methods

Detailed protocols are provided in *SI Appendix* for the adsorption experiments, liquid-state NMR measurements, solid-state X-ray diffraction data acquisition, and molecular modeling simulations. Briefly, we conducted two sets of adsorption with a natural MMT using either MgCl_2 or NaCl as the background electrolytes to probe the effects of divalent versus monovalent cations. The adsorption distribution coefficient or K_d values were obtained from single-compound experiments with each of the 10 biomolecules investigated or from mixed-compound experiments with a mixture containing all 10 biomolecules

simultaneously. The concentration of each biomolecule in solution was quantified by ^1H -NMR experiments using a Bruker NOE 600 MHz spectrometer equipped with a QCI-F cryoprobe. The layer-to-layer distance, or d_{001} , of the MMT slurries was captured without and with adsorbed biomolecule using a Bruker D8 Advance powder X-ray diffractometer under constant temperature and relative humidity. For molecular modeling of the adsorbate conformations, we employed the condensed-phase optimized molecular potentials for atomistic simulation studies (COMPASS, version II) force field with the Ewald method for electrostatic interactions and atom-based method for Van der Waals interactions. For the partial charges and bonding parametrization of the biomolecules, we used the chemistry at Harvard macromolecular mechanics (CHARMM) force field or density-functional theory in cases where the force field parameterization was inadequate; the CLAYFF force field was used for the clay structures. Model adsorbate conformations were obtained by performing first a Monte Carlo search of initial energy-minimized and geometry-optimized configurations of biomolecule adsorptives at water–clay interfaces, followed by molecular dynamics equilibration to obtain the final thermodynamically optimized adsorbate structures used for analysis of the biomolecule–clay complexes. Separate molecular dynamics simulations were conducted to map out the diffusion trajectory of each biomolecule from the bulk solution to the water–clay interface.

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

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1. C. Le Quere *et al.*, *Global Carbon Budget 2017* (Earth System Science Data, 2018), **vol. 10**, pp. 405–448.
2. J. Lehmann, M. Kleber, The contentious nature of soil organic matter. *Nature* **528**, 60–68 (2015).
3. K. Georgiou *et al.*, Global stocks and capacity of mineral-associated soil organic carbon. *Nat. Commun.* **13**, 3797 (2022).
4. J. D. Hemingway *et al.*, Mineral protection regulates long-term global preservation of natural organic carbon. *Nature* **570**, 228–+ (2019).
5. A. Ito, R. Wagai, Global distribution of clay-size minerals on land surface for biogeochemical and climatological studies. *Sci. Data* **4**, 170103 (2017).
6. G. Sposito, *The Chemistry of Soils* (Oxford University Press, 2008).
7. M. L. Whittaker *et al.*, Ion exchange selectivity in clay is controlled by nanoscale chemical-mechanical coupling. *Proc. Natl. Acad. Sci. U.S.A.* **116**, 22052–22057 (2019).
8. R. Sutton, G. Sposito, Molecular simulation of humic substance–Ca–montmorillonite complexes. *Geochim. Cosmochim. Acta* **70**, 3566–3581 (2006).
9. B. K. G. Theng, K. R. Tate, P. Beckerheidmann, Towards establishing the age, location, and identity of the inert soil organic-matter of a spodosol. *Zeitschrift Fur Pflanzenernahrung Und Bodenkunde* **155**, 181–184 (1992).
10. J. Gao *et al.*, Competition and surface conditioning alter the adsorption of phenolic and amino acids on soil minerals. *Eur. J. Soil Sci.* **68**, 667–677 (2017).
11. J. Gao *et al.*, Organic matter coatings of soil minerals affect adsorptive interactions with phenolic and amino acids. *Eur. J. Soil Sci.* **69**, 613–624 (2018).
12. M. Kleber *et al.*, Dynamic interactions at the mineral–organic matter interface. *Nat. Rev. Earth Environ.* **2**, 402–421 (2021).
13. M. Jaber *et al.*, Selectivities in adsorption and peptidic condensation in the (arginine and glutamic acid)/montmorillonite clay system. *J. Phys. Chem. C* **118**, 25447–25455 (2014).
14. C. L. Peng, Y. H. Zhong, F. F. Min, Adsorption of alkylamine cations on montmorillonite (001) surface: A density functional theory study. *Appl. Clay Sci.* **152**, 249–258 (2018).
15. P. J. Mitchell *et al.*, Solution-state NMR investigation of the sorptive fractionation of dissolved organic matter by alkaline mineral soils. *Environ. Chem.* **10**, 333–340 (2013).
16. X. F. Wang *et al.*, Cation bridging studied by specular neutron reflection. *Langmuir* **29**, 5520–5527 (2013).
17. X. Sheng *et al.*, Metal cation saturation on montmorillonites facilitates the adsorption of DNA via cation bridging. *Chemosphere* **235**, 670–678 (2019).
18. H. Y. Xie *et al.*, Charge-dependent regulation in DNA adsorption on 2D clay minerals. *Sci. Rep.* **9**, 6808 (2019).
19. L. H. de Oliveira *et al.*, When RNA meets montmorillonite: Influence of the pH and divalent cations. *Appl. Clay Sci.* **214**, 106234 (2021).
20. J. Wang, P. Somasundaran, Study of galactomannose interaction with solids using AFM, IR and allied techniques. *J. Colloid Interface Sci.* **309**, 373–383 (2007).
21. M. Kleber *et al.*, Mineral–organic associations: Formation, properties, and relevance in soil environments. *Adv. Agron.* **130**, 1–140 (2015).
22. G. S. Liu *et al.*, Adsorption of polysaccharide onto talc. *Miner. Eng.* **19**, 147–153 (2006).
23. L. F. Han *et al.*, Different mechanisms driving the preferential adsorption of dissolved organic matter by goethite and montmorillonite. *Chem. Geol.* **585**, 120560 (2021).
24. B. H. Gu *et al.*, Competitive adsorption, displacement, and transport of organic matter on iron oxide: II. Displacement and transport. *Geochim. Cosmochim. Acta* **60**, 2977–2992 (1996).
25. K. Sanderson, Lignocellulose: A chewy problem. *Nature* **474**, S12–S14 (2011).
26. A. Gunina, Y. Kuzakov, Sugars in soil and sweets for microorganisms: Review of origin, content, composition and fate. *Soil Biol. Biochem.* **90**, 87–100 (2015).
27. F. Q. Dong *et al.*, Spectroscopic evidence and molecular simulation investigation of the bonding interaction between lysine and montmorillonite: Implications for the distribution of soil organic nitrogen. *Appl. Clay Sci.* **159**, 3–9 (2018).
28. B. J. Smith *et al.*, Origins of saccharide-dependent hydration at aluminate, silicate, and aluminosilicate surfaces. *Proc. Natl. Acad. Sci. U.S.A.* **108**, 8949–8954 (2011).
29. L. Aristilde *et al.*, Sugar-influenced water diffusion, interaction, and retention in clay interlayer nanopores probed by theoretical simulations and experimental spectroscopies. *Adv. Water Resour.* **106**, 24–38 (2017).
30. J. N. Nkoh *et al.*, Adsorption of amino acids by montmorillonite and gibbsite: Adsorption isotherms and spectroscopic analysis. *Appl. Clay Sci.* **219**, 106437 (2022).
31. A. M. Ahmat, T. Thiebault, R. Guegan, Phenolic acids interactions with clay minerals: A spotlight on the adsorption mechanisms of Gallic Acid onto montmorillonite. *Appl. Clay Sci.* **180**, 105188 (2019).
32. Y. L. Yang *et al.*, Adsorption of lysine on Na–montmorillonite and competition with Ca^{2+} : A combined XRD and ATR-FTIR study. *Langmuir* **32**, 4746–4754 (2016).
33. N. Kitada, T. Yokoyama, S. Nakashima, In situ ATR-IR investigation of L-lysine adsorption on montmorillonite. *J. Colloid Interface Sci.* **338**, 395–401 (2009).
34. A. Parbhakar *et al.*, Adsorption of L-lysine on montmorillonite. *Colloids Surfaces a-Physicochem. Eng. Aspects* **307**, 142–149 (2007).
35. S. D. Zhu *et al.*, Adsorption and desorption of Pb(II) on L-lysine modified montmorillonite and the simulation of interlayer structure. *Appl. Clay Sci.* **169**, 40–47 (2019).
36. S. Yeamin *et al.*, Influence of mineral characteristics on the retention of low molecular weight organic compounds: A batch sorption-desorption and ATR-FTIR study. *J. Colloid Interface Sci.* **432**, 246–257 (2014).
37. A. M. Davis, G. Joannis, L. Tribe, Molecular Modeling of polymer–clay nanocomposite precursors: Lysine in montmorillonite interlayers. *J. Comput. Chem.* **29**, 983–987 (2008).

38. Y. Artioli, "Adsorption" in *Encyclopedia of Ecology*, S. E. Jorgensen, B. D. Fath, Eds. (Academic Press, 2008), pp. 60–65.
39. C. Boahen *et al.*, Adsorption of heavy metals from mine wastewater using amino-acid modified Montmorillonite. *Sustain. Environ.* **9**, 2152590 (2023).
40. L. Aristilde *et al.*, Enhanced interlayer trapping of a tetracycline antibiotic within montmorillonite layers in the presence of Ca and Mg. *J. Colloid Interface Sci.* **464**, 153–159 (2016).
41. J. Lindholm, J. F. Boily, M. Holmboe, Deconvolution of smectite hydration isotherms. *ACS Earth Space Chem.* **3**, 2490–2498 (2019).
42. E. Orucoglu *et al.*, Competitive adsorption processes at clay mineral surfaces: A coupled experimental and modeling approach. *ACS Earth Space Chem.* **6**, 144–159 (2021).
43. M. M. Fernandes, B. Baeyens, Cation exchange and surface complexation of lead on montmorillonite and illite including competitive adsorption effects. *Appl. Geochem.* **100**, 190–202 (2019).
44. L. M. Wu, L. B. Liao, G. C. Lv, Influence of interlayer cations on organic intercalation of montmorillonite. *J. Colloid Interface Sci.* **454**, 1–7 (2015).
45. L. M. Wu, S. Y. Cao, G. C. Lv, Influence of energy state of montmorillonite interlayer cations on organic intercalation. *Adv. Mater. Sci. Eng.* **2018**, 454 (2018).
46. S. Yang *et al.*, Competitive adsorption of metal ions at smectite/water interfaces: Mechanistic aspects, and impacts of co-ions, charge densities, and charge locations. *J. Phys. Chem. C* **124**, 1500–1510 (2020).
47. Y. Li, X. J. Wang, J. F. Wang, Cation exchange, interlayer spacing, and thermal analysis of Na/Ca-montmorillonite modified with alkaline and alkaline earth metal ions. *J. Thermal Anal. Calorimetry* **110**, 1199–1206 (2012).
48. X. B. Xing *et al.*, The binding energy between the interlayer cations and montmorillonite layers and its influence on Pb²⁺ adsorption. *Appl. Clay Sci.* **112**, 117–122 (2015).
49. Z. F. Chang *et al.*, Organo-mineral complexes protect condensed organic matter as revealed by benzene-polycarboxylic acids. *Environ. Pollution* **260**, 113977 (2020).
50. K. Fang *et al.*, Mineral and climatic controls over soil organic matter stability across the tibetan alpine permafrost region. *Global Biogeochem. Cycles* **35**, e2021GB007118 (2021).
51. Y. B. Zhang *et al.*, Interfacial reaction between humic acid and Ca-Montmorillonite: Application in the preparation of a novel pellet binder. *Appl. Clay Sci.* **180**, 105177 (2019).
52. I. A. Shabtai *et al.*, Calcium promotes persistent soil organic matter by altering microbial transformation of plant litter. *Nat. Commun.* **14**, 6609 (2023).
53. F. Ruiz *et al.*, Adsorption and thermal stability of dissolved organic matter on Ca- and Mg-exchanged montmorillonite: Implications for persistence in soils and sediments. *Chem. Geol.* **643**, 121813 (2023).
54. M. C. Rowley *et al.*, Association between soil organic carbon and calcium in acidic grassland soils from Point Reyes National Seashore, CA. *Biogeochemistry* **165**, 1–21 (2023).
55. M. C. Rowley, S. Grand, É. P. Verrecchia, Calcium-mediated stabilisation of soil organic carbon. *Biogeochemistry* **137**, 27–49 (2018).
56. L. Xie *et al.*, Probing the intermolecular interaction mechanisms between humic acid and different substrates with implications for its adsorption and removal in water treatment. *Water Res.* **176**, 115766 (2020).
57. L. Y. Wang *et al.*, Sorption of DONs onto clay minerals in single-solute and multi-solute systems: Implications for DONs mobility in the vadose zone and leachability into groundwater. *Sci. Total Environ.* **712**, 135502 (2020).
58. K. A. Block *et al.*, Exfoliation and intercalation of montmorillonite by small peptides. *Appl. Clay Sci.* **107**, 173–181 (2015).
59. W. E. Dubbin, J. P. Vetterlein, J. L. Jonsson, Fatty acids promote fulvic acid intercalation by montmorillonite. *Appl. Clay Sci.* **97–98**, 53–61 (2014).
60. K. S. Kim *et al.*, Massive intercalation of urea in montmorillonite. *Soil Sci. Soc. Am. J.* **75**, 2361–2366 (2011).
61. A. Naidja, A. Violante, P. M. Huang, Adsorption of tyrosinase onto montmorillonite as influenced by hydroxylaluminum coatings. *Clays Clay Miner.* **43**, 647–655 (1995).
62. D. R. Katti *et al.*, Mechanical properties of the sodium montmorillonite interlayer intercalated with amino acids. *Biomacromolecules* **6**, 3276–3282 (2005).
63. F. G. Mendonca *et al.*, Use of montmorillonite to recover carboxylic acids from aqueous medium. *Separat. Purif. Technol.* **229**, 115751 (2019).
64. H. Kostelnikova, P. Praus, M. Turicova, Adsorption of phenol and aniline by original and quaternary ammonium salts-modified montmorillonite. *Acta Geodynam. Geomater.* **5**, 83–88 (2008).
65. F. Demir *et al.*, Amino acid intercalated montmorillonite: Electrochemical biosensing applications. *RSC Adv.* **4**, 50107–50113 (2014).
66. M. von Lutzow *et al.*, Stabilization of organic matter in temperate soils: Mechanisms and their relevance under different soil conditions—a review. *Eur. J. Soil Sci.* **57**, 426–445 (2006).
67. G. Simonetti *et al.*, Soil porosity in physically separated fractions and its role in SOC protection. *J. Soils Sediments* **17**, 70–84 (2017).